IAS PRELIMS/MAINS

Ecology & Environment

A COMPREHENSIVE COVERAGE

MODULE - 24

WATER POLLUTION -

WATER POLLUTION DEFINITION

- It is the presence of unwanted substances in a water body such as organic, in-organic, radioactive substances and heat which degrade the quality of water.

EUTROPHICATION

- **Definition:** Eutrophication is the process of the addition of nutrients such as nitrates and phosphates naturally or artificially into the water body.
- It is fertilizing the aquatic ecosystem.
- Algal blooms are the consequence of Eutrophication.

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WATER POLLUTION

ALGAL BLOOM

Phytoplankton (algae) thrive on the excess nutrients and their population explosion covers the entire surface of the water body in a thick layer. This condition is known as algal bloom.

HARMFUL EFFECTS OF ALGAL BLOOM

1 DEPLETION OF OXYGEN FROM THE WATER BODY

- Algal blooms restrict the dissolution of atmospheric oxygen into the water body.
- Oxygen in aquatic ecosystem is also replenished by photosynthetic aquatic plants. Algal Blooms restrict the penetration of sunlight which prohibits them from carrying out photosynthesis resulting in death of aquatic plants, and hence restricts the replenishment of oxygen.
- The oxygen level is already depleted due to the population explosion of phytoplankton. Phytoplankton is photosynthetic during day time adding oxygen to aquatic ecosystem. But during nights, they consume far more oxygen as they respire.
- Algal blooms accelerate the rate of oxygen depletion as the population of phytoplankton is very high.
- The new anaerobic conditions created promote growth of bacteria such as Clostridium botulinum which produces toxins deadly to aquatic organisms, birds and human beings.

HARMFUL EFFECTS OF ALGAL BLOOM

2 DEATH OF AQUATIC ORGANISMS AND DECREASE IN THE BIO-DIVERSITY OF THE WATER BODY

- The primary consumers like small fish are killed due to oxygen deprivation caused by algal blooms.
- Death of primary consumers adversely affects the food chain and leads to the destruction of higher life forms.
- Further, more oxygen is taken up by microorganisms during the decomposition process of dead algae, plants and fishes. Due to reduced oxygen level, the remaining fishes and other aquatic organisms also die. All this eventually leads to degradation of the entire aquatic ecosystem.

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HARMFUL EFFECTS OF ALGAL BLOOM

3 OTHER HARMFUL EFFECTS

- **Loss of fresh water lakes** - Algal Blooms eventually create detritus layers in the bottom of water bodies and reduces the depth of the water body. Eventually the water body is reduced into marsh land.
- **New species invasion** - Eutrophication may cause some invasive species such as water hyacinth (Read further about Water Hyacinth in this Chapter) to grow abundantly and destroy the ecosystem.
- **Toxicity** - Some algal blooms release neuro & hepatotoxins which can kill aquatic organisms & pose threat to humans.
- **Loss of coral reefs** - Degradation of the aquatic ecosystem leads to the loss of coral reefs.
- **Affects navigation** - Algal Blooms affect navigation.

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HARMFUL EFFECTS OF ALGAL BLOOM

3 OTHER HARMFUL EFFECTS

- Eutrophication occurs naturally due to deposition of nutrients carried by flood waters. It takes over centuries for Eutrophication to occur naturally.
- Similar nutrient enrichment of water bodies at an accelerated rate is caused by human activities (discharge of wastewaters from houses or agricultural runoff etc.) and the resulting phenomenon is known as 'cultural eutrophication' and it takes only years.
- **Mitigation Measures** - Treating Industrial effluents and domestic sewage; Constructing Riparian buffers.

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HARMFUL ALGAL BLOOMS

- Most algal blooms are not harmful but some produce toxins that affect fish, birds, marine mammals and humans.
- The toxins may also make the surrounding air difficult to breathe. These are known as Harmful Algal Blooms (HABs).
- Harmful Algal Blooms are considered an environmental hazard because these events can make people sick when contaminated fish are eaten, or when people breathe toxins near the beach.

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